MEMBRAPOR
ELECTROCHEMICAL GAS SENSORS

Application Note MEM1:

Basics of Electrochemical Gas Sensor

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Inhalt

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1. General

MEMBRAPOR produces electrochemical gas sensors for fixed and portable gas detection equipment. It is expected that users of MEMBRAPOR products are professionals in the field of gas analysis and that they know their responsibility and their local regulations.

This application note is about **electrochemical** gas sensors for toxic gases and serves as a guidance. This documentation is not claimed to be complete and MEMBRAPOR does not accept a liability for any consequential losses, injury or damage resulting from the use of this document or the information contained within it.

2. Usage of our sensors

WARNING: SOLDERING OR CRIMPING TO PINS WILL RENDER YOUR WARRANTY VOID.

- Never solder connectors directly onto sensor pins. Connection must be made via a PCB mounting socket. To connect a sensor with wires, use the Slim- or Standard-size which have soldering tags.
- Do not bend or twist soldering tags (Slim- and Standard-size). Order sensors without soldering tags if you do not need them.
- If there is a short-cutting spring, remove it before plugging the sensor in.
- Do not remove the white anti-condensation membrane.
- Sensors must not be exposed to temperatures, humidities and pressures outside the specified range in the respective data sheets.

- Sensors should not be exposed to organic vapor as it may influence the baseline or even cause physical damage to the sensor body (see chapter 18.1).
- At the end of product's life, sensors can be returned to MEMBRAPOR for professional and environment-friendly disposal.

General: The sensors are not suitable for temperatures over 70 °C and for use in vacuum conditions (max. pressure differential of 100 mbar below ambient pressure).

3. Intrinsic Safety Considerations

An electrochemical sensor is a cell which produces small currents and voltages and is unable to store large quantities of energy. The sensor’s current increases linearly over the recommended operating range of the gas concentration and is measured as the sensor output:

sensor sensitivity [nA/ppm] x gas concentration [ppm] = output signal [nA]

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Maximum current in normal operation:	< 0.2 mA
Maximum short circuit current:	< 0.5 A
Maximum voltage in normal operation:	< 1.2 V
Maximum open circuit voltage:	< 1.2 V

Electrochemical gas sensors for toxic gases comply to IEC 60079-11, 5.6. This considers the sensors to be a simple apparatus not exceeding a maximal energy of 1.5 V, 100 mA and 25 mW.

4. Product Safety

A MEMBRAPOR gas sensor is not considered hazardous. Should the housing be damaged, the electrolyte inside the sensor may leak out. Exposure to the sensor electrolyte (typically diluted sulfuric or phosphorous acid), is the only risk that may potentially prove hazardous to health. In the event of skin contact, rinse with plenty of water and seek medical advice. The MSDS document can be downloaded from our website.

5. Storage and Transport

The sensors must be stored in their original packaging as delivered by Membrapor. We recommend a general storage temperature range between -10°C and 30°C and at a relative humidity of 20 to 80 %. At such conditions the natural aging effects are slowed down. Please note that the sensor's lifetime starts with its production. Therefore, we recommend using them immediately. The exact storage conditions are described in the sensor data sheets.

Do not store sensors together with organic solvents or flammable liquids.

6. Sensor Housing

MEMBRAPOR offers different sensor housings, all made of polycarbonate. The housings determine different sensor features such as sensitivity, filter and electrolyte capacity or resistance to poisoning effects.

Membrapor housing	Industry size
Compact	Conforms to 7-Series
Vantage	Conforms to 5-Series
Miniature	Conforms to 4-Series
Slim	Conforms to 3-Series
Standard	Pinlayout as 3-Series. Larger electrolyte volume

Table 1: Overview of sensor housing.

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The exact dimensions and weights are depicted in the corresponding datasheets.

7. Connecting the Sensor

Electrochemical sensors have either 2, 3 or 4 pins at the bottom for the electrical and mechanical contact. The Slim- and Standard-size sensors have additionally soldering tags, which can be used for the electrical contact. The PCB should have sockets where the sensor with its pins can be plugged in.
Sockets with a clip give generally a safer electrical connection with the pin. They should be considered in case of vibrations or frequent plugging.

The pin diameters are:

- Compact, Slim, Standard: 1 mm (0.04 in)
- Miniature, Vantage: 1.5 mm (0.06 in)

8. Labelling

The label on MEMBRAPOR gas sensors includes the sensor type, the serial number and a 2D-Code which contains both information.

8.1. Sensor type

Every sensor type is described with a logical product code:

[Gas] / [Size Sensor] [*Option* Filter] [*Option* Variation] - [Measurement Range] - [*Option* 2 *or* 4 Electrodes] - [*Option* Slim *if S-size*]

Examples:

CH2O/C-10	→	For Formaldehyde, Compact-size, for 0-10 ppm
NO/SF-2000-S	→	For Nitric Oxide, Slim-size, with filter, for 0-2000 ppm
NH3/MR-100	→	For Ammonia, with fast response, for 0-100 ppm

8.2. Serial Number and Date of Manufacture

With the individual serial number, MEMBRAPOR can trace all test results and production data, including the whole supply chain.

The date of production is encoded in the first 4 digits: yyCW.

yy: The year (Last 2 digits, e.g. 14 means the year 2014)
CW: The calendar week in that year.

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8.3. 2D-Code

On the white label is a two-dimensional matrix bar code printed. It is encoded in "DataMatrix Code", ECC type: 200. It includes the serial number and the sensor type. This two information are separated with a space.

9. Sensor principle

9.1. Electrochemical Gas Sensors in General

Electrochemical sensors operate by reacting with the analyte and producing an electrical signal. Most electrochemical gas sensors are amperometric sensors, generating a current that is linearly proportional to the gas concentration. The principle behind amperometric sensors is the measurement of the current-potential relationship in an electrochemical cell where equilibrium is not established. The current is quantitatively related to the rate of the electrolytic process at the sensing electrode (also known as working electrode) whose potential commonly is kept constant using another electrode (the so-called reference electrode).

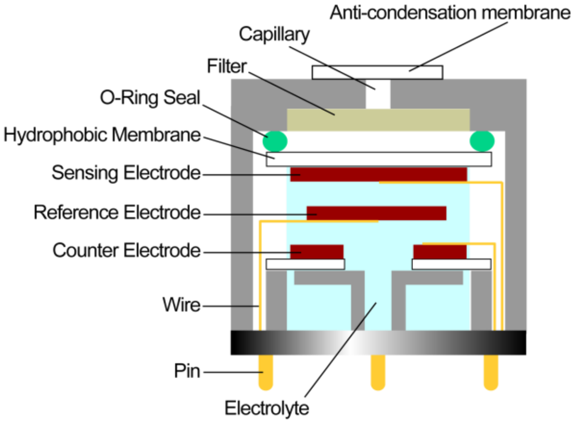

Figure 1: Working principle of an electrochemical gas sensor.

9.2. Working principle

A MEMBRAPOR electrochemical gas sensor works as follows: Target gas molecules that encounter the sensor first pass an anti-condensation membrane which serves also as a protection against dust. Then the gas molecules diffuse through a capillary, potentially through a subsequent filter, and then through a hydrophobic membrane to reach the surface of the sensing electrode. There the molecules are immediately oxidized or reduced, consequently producing or consuming electrons, and thus generating an electric current.

It is important to note that with this approach the amount of gas molecules entering the sensor is limited by the diffusion through the capillaries. By optimizing the pathway, in accordance with the desired measurement range, an adequate electrical signal is obtained. The design of the sensing electrode is crucial to both achieve a

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high reactivity towards the target gas and to inhibit undesired responses to interfering gases. It involves a system of three phases: solid, liquid and gaseous, and all are involved in the chemical recognition of the analyte gas. MEMBRAPOR is passionately dedicated to tailor this system and obtain high-performance gas sensors.

The electrochemical cell is completed by the so-called counter electrode which balances the reaction at the sensing electrode. The ionic current between the counter and sensing electrode is transported by the electrolyte inside the sensor body, whereas the current path is provided through wires terminated with pin connectors.
Commonly, a third electrode is included in an electrochemical sensor (3-electrode sensor). The so-called reference electrode serves to maintain the potential of the sensing electrode at a fixed value. For this purpose and generally for the operation of an electrochemical sensor a potentiostatic circuit is needed.

9.3. Sensor Signal

The output signal of a MEMBRAPOR gas sensor corresponds to the concentration of a gas rather than to its partial pressure. Hence, it is possible to use a MEMBRAPOR sensor at different altitudes or even underground, independent at which atmospheric pressure the device was calibrated.
A deeper and scientific explanation of the sensor output and the pressure dependence can be found in the document MEM4.

10. 3-Electrode Sensors

To operate an electrochemical sensor a control circuitry is required, referred to as the potentiostatic circuit. For a 3-electrode sensor the main purpose is to maintain a voltage between the reference electrode (Ref) and the sensing electrode (Sens, also known as working electrode) to control the electrochemical reaction and to deliver an output signal proportional to the current produced by the sensor.
The Sens responds to the target gas, either oxidizing or reducing the gas, creating a current flow that is proportional to the gas concentration. This current must be supplied to the sensor through the counter electrode (Cnt).
At the Cnt the opposite redox reaction takes place, completing the circuit with the Sens. The potential of the Cnt is allowed to float. When gas is detected, the cell current rises and the Cnt polarizes with respect to the Ref. The potential of the Cnt is not important, if the circuit can provide sufficient voltage and current to maintain the correct potential of the Sens.

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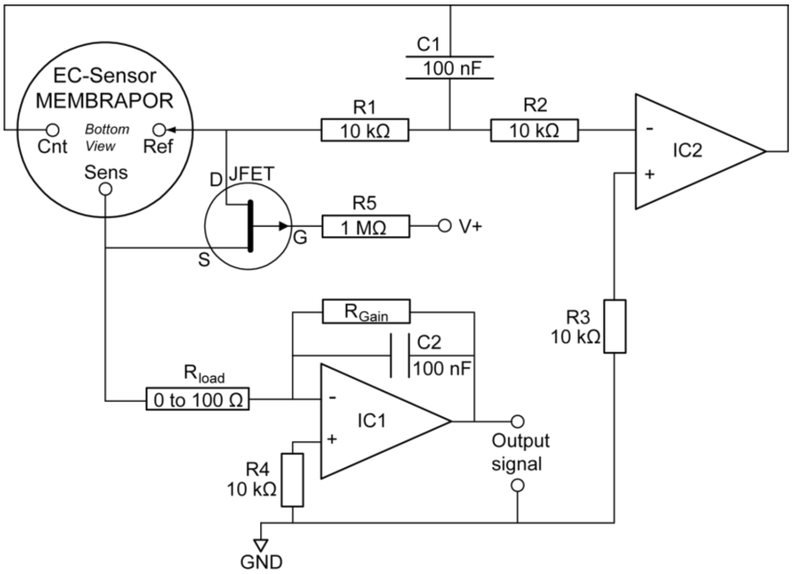

Figure 2: Schematic diagram of the electronic circuit for 3-Electrode sensors. This is only for illustration.

The measuring circuit for the electrochemical sensor is a single stage operational amplifier IC1 in a transimpedance configuration. The sensor current is reflected across R_{GAIN}, generating an output voltage relative to the virtual earth GND. C2 reduces high frequency noise. The load resistor R_{load} is a compromise between fastest response time and best signal-to-noise ratio. The recommended value is given in the sensor data sheet.

The control operational amplifier IC2 provides the current to the Cnt to balance the current required by the Sens. The inverting input into IC2 is connected to the Ref and must not draw any significant current from it.

The JFET is shorting the Ref to the circuit common when the circuit power is switched off. This ensures that the Sens is maintained at the same potential as the Ref. This state for unbiased sensors ensures that the sensor is ready immediately when switched on.

Sensors which need a bias voltage have not to be shorted. In opposite, it is recommended to hold the voltage (mostly +300 mV) all the time. This can be realized with a button cell during

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power off.

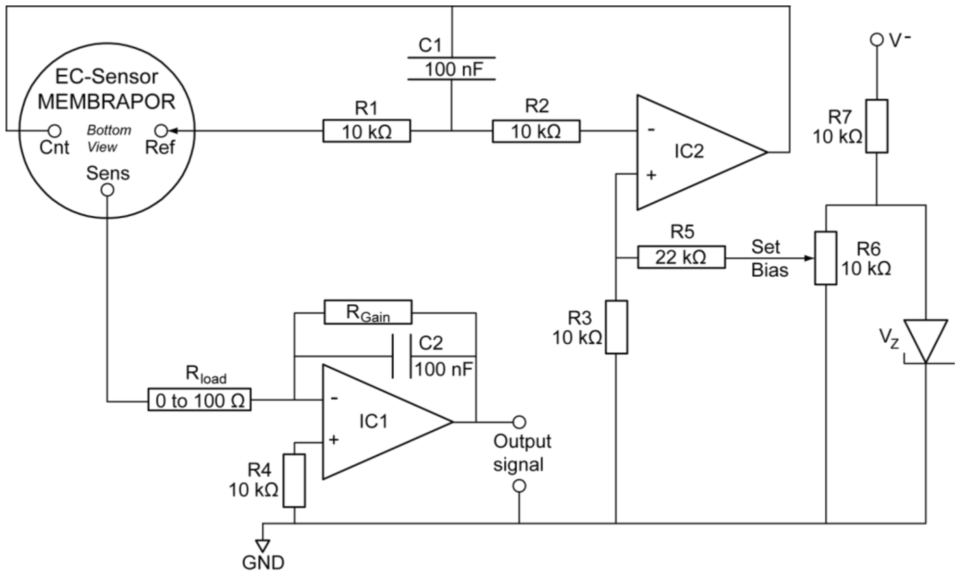

Figure 3: Schematic diagram of the electronic circuit for biased 3-Electrode sensors.

11. 2-Electrode Sensors

A 2-Electrode sensor has the disadvantage that the potential of the sensing electrode shifts away. This occurs mainly when it is exposed to the target gas. Therefore, it is not recommended to use a 2-electrode sensor for continuous measurement. Other disadvantages are stronger temperature dependence and lower signal output, compared to 3 or 4-electrode sensors.

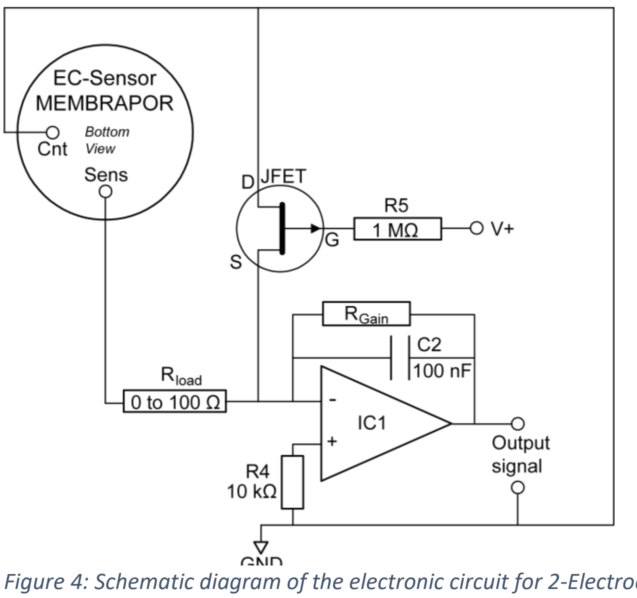

Figure 4: Schematic diagram of the electronic circuit for 2-Electrode sensors.

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12. 4-Electrode Sensors

4-electrode sensors are the most sophisticated electrochemical sensors. They have an additional electrode, the so-called auxiliary electrode (Aux), that gives additional information about the state of the sensor. MEMBRAPOR produces 2 different kinds of 4-electrode sensors:

- Hydrogen-compensated sensors
- L-type sensors for low concentrations

These are completely different types of sensors, where the Aux adopts different functions. However, the basic potentiostatic circuit is the same:

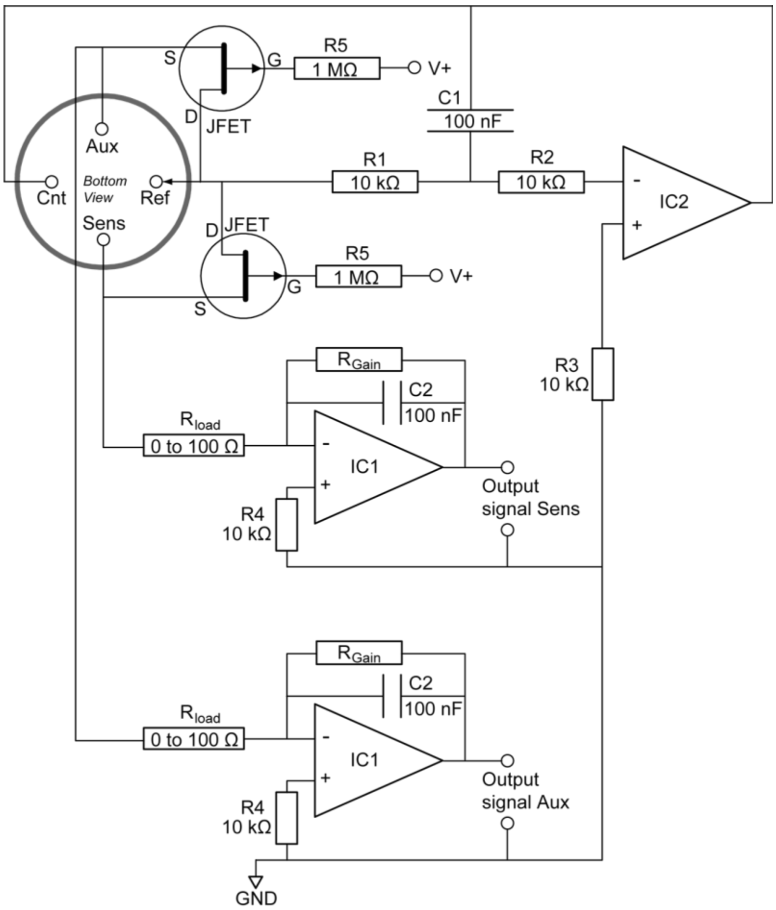

Figure 5: Schematic diagram of the electronic circuit for 4-Electrode-sensor

12.1. Hydrogen-compensated Sensors

This type of MEMBRAPOR 4-electrode sensor allows to compensate for an interfering gas which is not possible to discriminate with filters or other techniques. The CO/VF-1000-4E is a very popular gas sensor in flue gas/stack gas monitoring and an example of a hydrogen-compensated carbon monoxide (CO) sensor. While both the oxidation of CO and interfering hydrogen (H_2) occur at the sensing electrode, the signal at the auxiliary electrode is almost exclusively due to H_2. With

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two signals caused by two gases the concentrations can easily be calculated and thus, a highly accurate CO concentration is obtained in applications where 3-electrode sensors would fail.

12.2. L-Type Sensors for low concentrations

This type of MEMBRAPOR sensor allows to decrease the detection limit and increase the resolution compared to 3-electrode sensors. The signal of the auxiliary electrode is used to correct for the following effects:

- Drift of the baseline caused by changes in temperature
- Drift of the baseline caused by interference

Not corrected are cross-sensitivities due to interfering gases.

For further information on 4-Electrode sensors please revise our application note MEM6.

13. Gas Sampling System

13.1. Positioning of the Sensor

The orientation of a MEMBRAPOR sensor has no influence on its operation. Do avoid pressure-related effects and bulk flow MEMBRAPOR recommends installing the sensor perpendicular to the gas flow. In this case diffusion works optimally.
In a device which is equipped with a pump, preferably the sensor is installed upstream of the pump. This minimizes any loss of analyte gas and any effects resulting from shock pressures caused by membrane pumps.

13.2. Flow Rate

When using a pump or mass flow controllers, the flow rate of the gas sample must be high enough to avoid depletion of the analyte. MEMBRAPOR recommends a minimal flow rate of 150 ml/min. If a dependence of the flow rate is observed, then the flow must be increased or the cross-sectional area above the sensor must be reduced. It is generally recommended to have a low dead volume above the sensor top. Typical cases, where depletion occurs, are while measuring NO_2 gas and in general with highly sensitive sensors and low gas concentrations.

13.3. Discontinuous measurement

Some gases, like hydrogen sulfide (H_2S) or ammonia (NH_3), can affect the sensing capability of a sensor. Additionally, in some applications like biogas, there are

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additionally further sulfur containing compounds which can irreversibly poison the sensor. See chapter 18.

In such a situation the solution is to measure in discontinuous mode. In this manner the operation time will last more years. Conversely, the lifetime can be shortened to a few or even one month when measuring continuously.

A typical cycle in such a discontinuous mode is as follows:

- 3 min measurement
- 10 till 30 min purging with fresh air

Usually, the fresh air is obtained by tubing from the outside.

13.4. Oxygen

All gas sensors for reducing gases (CO, H_2, H_2S, NO...) require oxygen (O_2) to provide the counter-reaction at the counter electrode. A rule of thumb is that the O2-concentration in the environment needs to be at least 5x the full-scale analyte's concentration. Hence, a CO/CF-200 sensor requires at least 1000 ppm (or 0.1 Vol%) of O2 to operate smoothly.

But there are several possibilities to measure in a gas sample without O_2:

- Short time measurement of a few minutes. Before and after the measurement, the sensor is exposed to ambient air. In this way, it has enough O_2 dissolved in the electrolyte.
- Compact-, Slim- and Standard-size sensors have a little hole on the back side for O_2 access. These types can be front mounted to a pipe with the gas stream and get O_2 from ambient air from the back side.
- Discontinuous measurements. Gas sampling system with valves which switch between gas sample and ambient air periodically.

The same measures can be applied to gas samples with humidities outside of the specified range.

13.5. Humidity

General

Most electrochemical gas sensors contain a liquid, aqueous electrolyte which is in equilibrium with the humidity content of surrounding air. At dry conditions the electrolyte will lose water and gain water at high humidities. Most MEMBRAPOR sensors are specified to operate in the humidity range of 15 % to 90 % RH (see Figure 6). However, for a short time the sensor can be operated outside the humidity range.

Sensor Performance

All sensor performance data are monitored at laboratory conditions (room temperature, typical humidity 40 – 60 % R.H., 1013 mbar pressure) using MEMBRAPOR's analogue transmitters. The sensor data is not post-processed. The

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sensitivity of a MEMBRAPOR gas sensor is **independent of the humidity** over the specified range. An abrupt change in humidity can cause a short-term transient signal for some types of sensors. This effect is noted in the respective data sheets. Calibration with dry gas, when switching from ambient air, is such a situation. Thereby the first seconds of the sensor response are a superposition of this transient signal and the signal response to the change in gas concentration.

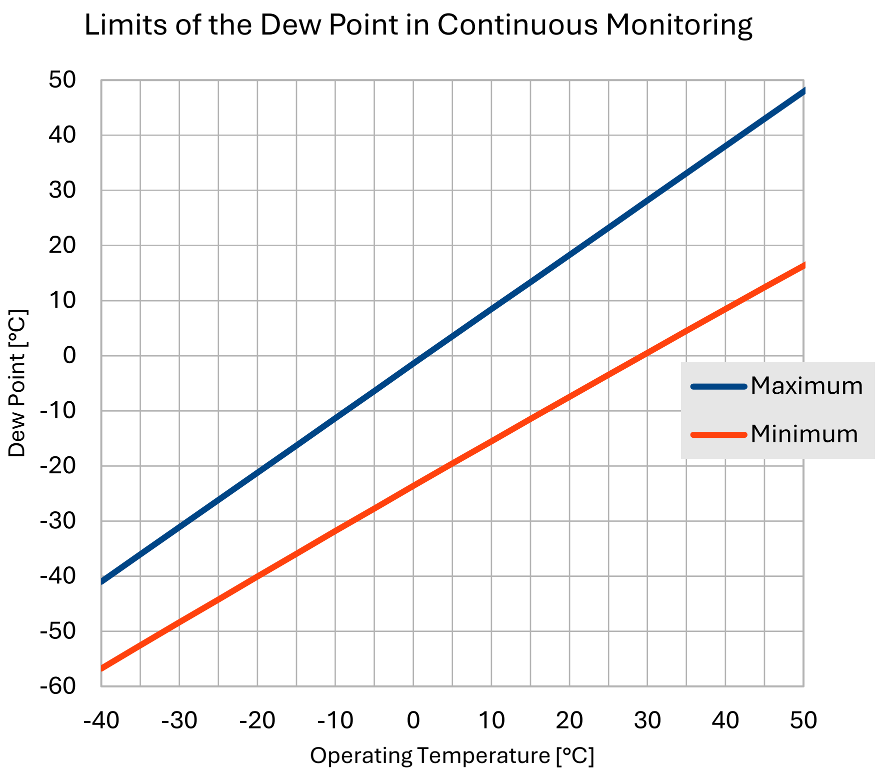

Figure 6: Allowed humidity range expressed as limits of the dew point of the gas sample.

Extreme Conditions of Humidity

Operation for extended periods outside the range of 15% to 90% relative humidity will invalidate the warranty.

At high relative humidity (R.H. > 90%), the liquid electrolyte uptakes water from the environment. At some point the electrolyte reservoir's volume is exceeded resulting in leakages. In contrary, at low relative humidity (R.H. < 15%), the electrolyte releases water to the atmosphere resulting in a stronger acid concentration and finally in corrosion of the sensor pins. Depending on the capillary surface area this phenomenon can take several weeks. **MEMBRAPOR also offers special sensor types in the Compact housing for dry environments (denoted with letter H) and for humid environments (denoted with letter T)**

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13.6. Temperature

The temperature range, where a sensor can be used, is specified in the data sheet. Generally, the maximum temperature is +60 °C. Above this limit the sealing in the sensor housing might be affected and the electrolyte could leak out. Additionally, inner electrical contacts might be corrupted, which would result in a slow response and signal loss.

Low temperatures are harmless. But for some gases the signals at -40 °C can be very low.

In the case of temperature gradients in a gas sampling system, attention has be paid to avoid condensation onto the sensor head. A formation of droplets on the anti-condensation membrane will affect the gas entry and therefore the sensor response.

13.7. Pressure

Generally, electrochemical gas sensors are meant to be operated at atmospheric pressure ±10 %. However, MEMBRAPOR gas sensors can be used in a broader pressure range. Ask the MEMBRAPOR technical support team for help.

13.8. Partial Pressure versus Concentration

The signal of MEMBRAPOR gas sensors and therefore the calibration is not affected by changes in total pressure, or only to a small degree. In opposite to other gas sensors, MEMBRAPOR sensors give a signal proportional to the concentration of the analyte gas and not to its partial pressure.

The scientific explanation can be obtained from MEMBRAPOR's technical support team.

In practice, an instrument with a MEMBRAPOR gas sensor can be used at any elevation or underground, without the need to recalibrate it.

14. Calibration

The performance of a sensor or the whole instrument, respectively, must be checked regularly with calibration gas. Generally a recalibration is necessary when the sensor ‘s sensitivity is below 50% of its initial value or if the response time has increased significantly.

Calibrate the device with the gas sensor at conditions most like the intended usage. Use a gas mixture representing the gas matrix in the application. This is especially important to avoid background gas effects as described in MEM3. Perform the span-calibration with the target gas. In some rare cases the cross-sensitivity to a different gas can be used.

It is uncritical to use completely dry calibration gas mixtures. Within the time of calibration, the sensor will not dry out. MEMBRAPOR gas sensors without a bias voltage are ready for calibration/measurement within 3 minutes after power on. Sensors requiring a bias voltage need between 24 to 72 hours for stabilization of the

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baseline and adaption to the environmental temperature. Start a calibration/measurement when the baseline is stable.

The calibration interval depends on number of factors including application, environmental conditions, regulations and accuracy requirements. The long-term output drift of a sensor is reduced with time at constant conditions.

15. Temperature compensation

The output signal (sensitivity) and the baseline of a sensor are temperature-dependent parameters. Generally, the sensitivity increases with increasing temperature and is at -40 °C much lower than at room temperature. Basically, this is given by the nature of the gas and its chemical reaction inside the sensor.

For some gases the baseline differs from zero, mostly at high temperatures and this especially with biased sensors.

It is highly recommended to acquire the temperature dependence curves with the whole instrument. The sampling system, the electronics, the interaction between the electronics and the sensor, all have a significant impact on the temperature dependence of the final measurement reading.

In application note MEM7 the procedure for temperature compensation is described in detail.

16. How to Check Sensor Functionality

The signal noise gives valuable information about the plugged-in sensor. Three cases can be distinguished: The signal noise

a. is too low, determined mainly by the circuit:
   The sensor is drying out or poisoned
b. is as low as if no sensor would be plugged in:
   The sensor has dried out completely or an electrical contact is lost
c. is too high: Leakage of the sensor electrolyte is occurring

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17. Gases and Lifetime

The lifetime of a sensor depends heavily on the application and the conditions (temperature, humidity, gas matrix). Table 2 showcases MEMBRAPOR's main target applications. The conditions and challenges described are based on experience and customer testimonies. The expected sensor lifetime is derived from the conditions but can generally deviate depending on the exact conditions.

Application	**Conditions**	**Challenge**	**Expected sensor lifetime**
Biogas and Landfill	Harsh	Catalyst poisons (sulfuric compounds, Siloxanes, see 18	Low
Flue gas and exhaust gas	Harsh	Complex gas matrix, filter capacity	Middle – high
Air quality monitoring	Mild	Temperature and humidity stability	High
Odor emissions	Mild	Temperature and humidity stability	High
Disinfection and sterilization processes	Mild	Cross-sensitives and possible false alarms	Very high
Semiconductor industry	Moderate	Detection of very low signal levels	Middle
Medical applications	Mild	Cross-sensitives and possible false alarms	Very high
Industrial safety and personal protection	Moderate	Cross-sensitives and possible false alarms	Middle
Food and beverage industry	Moderate	Cross-sensitivities	Middle
Agricultural industry	Moderate	Cross-sensitivities	Middle
Confined spaces	Mild	Cross-sensitives and possible false alarms	Very high

Table 2: Typical applications for MEMBRAPOR sensors.

Based on the different gas types Table 3 summarizes the typical lifetimes.

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Gases	Typical lifetime
AsH3, C2H4, CS2, H2S, H2O2, HCl, HCN, NH3, PH3, SiH4	A lifetime of >2 year can be expected for leak detection, discontinuous measurement (see chapter 13.3) or spot measurement. Continuous exposure, even at low concentrations, must be avoided.
Acid, Alc, CH2O, Cl2, ClO2, NO2, O3, SO2	A lifetime of 3 – 5 years can be expected in the case of continuous exposure at low concentrations or spot measurements over the whole measurement range.
ETO, H2, CO, NO, O2	A lifetime of 3 - 5 years can be expected even in the case of continuous exposure.
O2	A lifetime of 5+ years can be expected at continuous exposure.

Table 3: Overview lifetime for different gases.

The performance of a sensor should be verified with regular tests or calibrations. If a sensor has lost more than half of its initial output signal, it is recommended to replace it. In continuous monitoring the noise of the signal can be used to check the sensor functionality (see chapter 16 How to Check Sensor Functionality).

18. Observable Effects of Certain Gases

18.1. Volatile Organic Compounds (VOCs)

Volatile organic compounds (VOCs) should be avoided in applications where electrochemical sensors are used because they can either dissolve in the electrolyte (e.g. ethanol) or adsorb onto the sensor's housing (e.g. dichloromethane, methyl ethyl ketone). Both effects can result in shifts in the baseline and/or changes in the overall sensor performance (sensitivity, response time...). Conveniently, the sensor can recover under clean air after a few days, depending on the exposure concentration and duration. The only exception to the above rule is Alc and VOC sensors, where allowed target VOCs are specified in the respective data sheets.

18.2. Sulfur containing compounds

Compounds containing sulfur strongly adsorb onto the catalytic surface, temporarily inhibiting the sensing reaction. Moreover, many of these compounds decompose and form a solid barrier. With prolonged exposure such processes lead to an irreversible decrease in sensitivity. This effect is known as poisoning of a catalyst.

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Use sensors with filters (Option *F*) if such gases are present in the intended application.

18.3. Silicones

Silicon compounds are known catalyst poisons.

19. Selectivity and Cross-Sensitivity

MEMBRAPOR gas sensors belong to the most selective sensors. Still, some limitations are given due to the chemical nature of the reactions involved. **All cross-sensitivity data contained in the data sheets refer to laboratory measurement conditions**

19.1. SO2 and NO2

All sensors for sulfur dioxide (SO_2) show a significant cross-sensitivity to nitrogen dioxide (NO_2). NO_2 produces a signal in opposite direction to the SO_2 signal, as it is reduced and not oxidized as SO_2. There is no suitable filter material that at once blocks NO_2 and let's pass SO_2. Therefore, measuring SO_2 in applications where also NO_2 is present, e.g. in flue gases, one must account for such negative signals when designing the electronic circuitry for a SO_2 sensor. Additionally, a sensor for NO_2 is needed to calculate the correct SO_2 concentration from the superimposed signals.
NO_2 sensors do not show a significant cross-sensitivity to SO_2 at normal concentrations. But to measure NO_2 near the detection limit with high SO_2 background, the same correction must be done as described above.

19.2. NO and NO2

In an electrochemical sensor nitrogen monoxide (NO) is oxidized to nitrogen dioxide (NO_2) and NO_2 is reduced to NO. These reactions must be taken in account when both sensors are mounted in the same line of the gas sampling system.

19.3. CO and H2

All sensors for carbon monoxide (CO) have a cross-sensitivity to hydrogen (H_2). With the A-type CO sensors from MEMBRAPOR it is strongly reduced and might be sufficient for many applications. Full discrimination is achieved with hydrogen-compensated 4-electrode sensors that give two signals and allow the exact calculation of both the CO and H_2 concentrations.

19.4. Filter options

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Due to the target gases' chemical nature the application of filters is not always possible. Table 4 gives a short overview for which target gases, sensors with optional inboard filters are available and the main effect of the filter. More detailed information can always be found in the corresponding data sheet.

Gas	**Effect of filter**
CO	*Removes acidic gases*
CS2	*Removes acidic gases*
NO	*Removes effect of SO2*
NO2	*Removes O3*
SO2	*Removes H2S and HCl*

Table 4: Gas sensors with optional filter

19.5. Filter Capacity

In several applications the filter capacity is crucial for the long-term performance of the electrochemical sensors. It can often be the limiting factor for the sensor lifetime. We determine the filter capacity based on a standard experiment with SO2 and NO2 (typical flue gas mixture). Whenever there is a breakthrough of 5%, the filter is used up. The filter capacity is denoted in ppm·hours.

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